



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

UNIT TEST

14-03-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

PAPER-2

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2022

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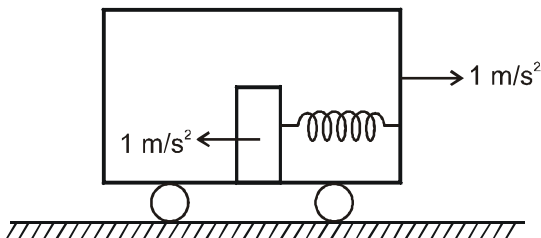
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PART 1 - PHYSICS

SECTION-I : (Maximum Marks : 80)

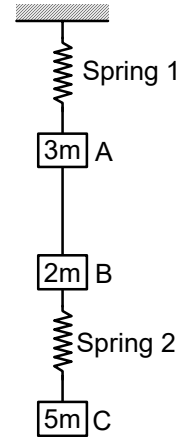
- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A block of mass 2 kg is placed inside a cart with a spring (spring constant = 100 N/m) attached with the block and the wall of the cart as shown. The cart is moving with a constant acceleration 1 m/s^2 with respect to the ground. At an instant the block is found to be accelerating with an acceleration 1 m/s^2 with respect to the cart when the spring is compressed by 0.1 m. The coefficient of kinetic friction between the block and the surface of the cart is : (Take $g = 10 \text{ m/sec}^2$)



(A) 0.2 (B) 0.7 (C) 0.5 (D) 0.6

2. In the given figure, the system is in equilibrium. If the string between A and B is cut, then choose the **INCORRECT** option

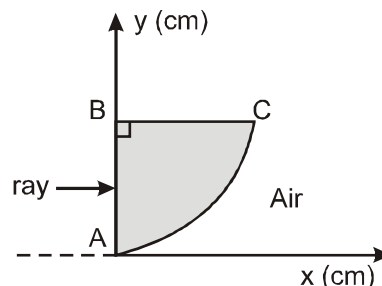


- (A) The instantaneous acceleration of A just after the string is cut is $\frac{7g}{3}$ upwards
- (B) The instantaneous acceleration of B just after the string is cut is $\frac{7g}{2}$ downwards
- (C) The instantaneous acceleration of C just after the string is cut is g downwards
- (D) The tension in the spring 1 just after the

string is cut is 10 mg

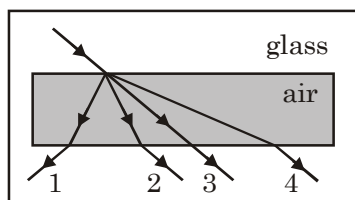
3. A point object is embedded in a glass slab of refractive $\mu = 1.5$. The point object is viewed normally through six faces of slab. Through five faces the object appears at 2cm from the surface while for sixth face it appears at 4cm from the surface. The volume of the slab is :
- (A) 108 cm^3
 (B) 546 cm^3
 (C) 324 cm^3
 (D) 144 cm^3
4. A point object is kept between a plane mirror and a concave mirror facing each other. The distance between the mirrors is 22.5 cm. Plane mirror is placed perpendicular to principal axis of concave mirror. The radius of curvature of the concave mirror is 20 cm. What should be the distance of the object from the concave mirror so that after two successive reflections the final image is formed on the object itself ? (Consider first reflection from concave mirror)
- (A) 5 cm
 (B) 15 cm
 (C) 10 cm
 (D) 7.5 cm

5. The cross-section ABC of an optical media of refractive index $\mu = \frac{5}{3}$ is as shown in figure. The curved surface is given by $y = x^2$. A ray of light travelling parallel to x-axis is incident normally on face AB. The minimum distance of the incident ray from point A to get refracted ray from surface AC is :-

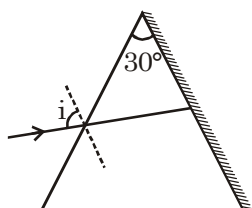


- (A) 1 cm (B) $\frac{2}{9}$ cm (C) $\frac{4}{9}$ cm (D) $\frac{2}{3}$ cm
6. A double convex lens forms a real image of an object on a fixed screen. Now the lens is given a constant velocity v_0 along its axis and away from the screen. For the purpose of forming the image always on the screen, the object is also required to be given an appropriate velocity.
- (A) If object size is half the image size then
 velocity of object = $\frac{3v_0}{2}$
- (B) If object size is double the image size then
 velocity of object = $\frac{v_0}{4}$
- (C) If object size is half the image size then
 velocity of object = $4v_0$
- (D) If object size is double the image size then
 velocity of object = $3v_0$

7. A laser beam propagating in glass (shown in white) hits the rectangular gap filled with air (shown in grey). Which line shown on the figure below represents correct path of the beam ?

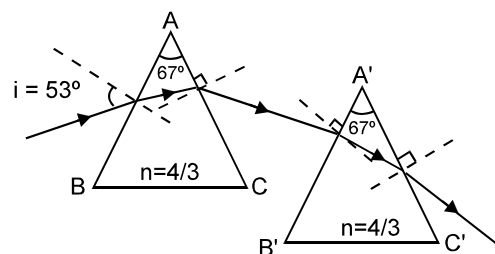


- (A) 1 (B) 2
(C) 3 (D) 4
8. A prism of refractive index $\sqrt{2}$ has a angle of prism 30° . One of the refracting surface of prism is silvered. Find the angle of incidence i for which the ray retraces its path :-



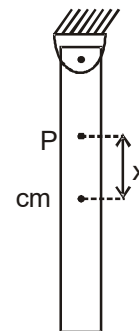
- (A) 30° (B) 60°
(C) 45° (D) 90°

9. A ray is incident on the first prism at an angle of incidence 53° as shown in the figure. The angle between side CA and B'A' for the net deviation by both the prisms to be double of the deviation produced by the first prism, will be :-



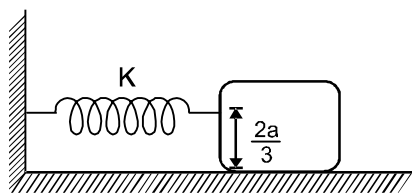
- (A) $2 \sin^{-1} \frac{2}{3}$ (B) $\sin^{-1} \frac{2}{3}$
(C) $\cos^{-1} \frac{2}{3}$ (D) $\sin^{-1} \frac{1}{3}$
10. A particle performing S.H.M. undergoes displacement of $\frac{A}{2}$ (where A = amplitude of S.H.M.) in one second. At $t = 0$ the particle was located at either extreme position or mean position. The time period of S.H.M. cannot be: (consider all possible cases)
- (A) 12s (B) 2.4
(C) 4s (D) 1.2s

11. A particle of mass m is performing linear simple harmonic motion. Its equilibrium is at $x = 0$, force constant is K and amplitude of SHM is A . The maximum power supplied by the restoring force to the particle during SHM will be:
- (A) $\frac{K^{\frac{3}{2}}A^2}{\sqrt{m}}$ (B) $\frac{2K^{\frac{3}{2}}A^2}{\sqrt{m}}$ (C) $\frac{K^{\frac{3}{2}}A^2}{3\sqrt{m}}$ (D) $\frac{K^{\frac{3}{2}}A^2}{2\sqrt{m}}$
12. A loaded vertical spring executes SHM with period of 4s. The difference between the kinetic energy and the potential energy of this system oscillates with a period of
- (A) 8 s (B) 1 s (C) 2s (D) 4s
13. The energy of a particle executing simple harmonic is given by $E = ax^2 + bv^2$ where ' x ' is the displacement from mean position $x = 0$ and v is the velocity of the particle at x then choose the **INCORRECT** statements.
- (A) amplitude of S.H.M is $\sqrt{\frac{E}{a}}$
- (B) Maximum velocity of the particle during S.H.M is $\sqrt{\frac{E}{b}}$
- (C) Time period of motion is $2\pi\sqrt{\frac{b}{a}}$
- (D) displacement of the particle is proportional to the velocity of the particle.
14. A particle moves along X axis such that its acceleration is given by $a = -\beta(x - 2)$. Where β is a positive constant and x is the position coordinate. The time period of oscillation (in sec) :-
- (A) π (B) 2π
- (C) $\frac{\pi}{\sqrt{\beta}}$ (D) $\frac{2\pi}{\sqrt{\beta}}$
15. A compound pendulum (a thin uniform rod of length 1m) is suspended from one of its end. When it is displaced slightly, it vibrates with time period T_0 . How many total number of points are on the rod about which if rod is hinged, it vibrates with same time period T_0 .



- (A) 4 (B) 2
- (C) 6 (D) infinity

16. Find maximum amplitude for safe SHM (block does not topple during SHM) of a uniform cubical block of side 'a' on a smooth horizontal floor as shown in figure (spring is massless) :

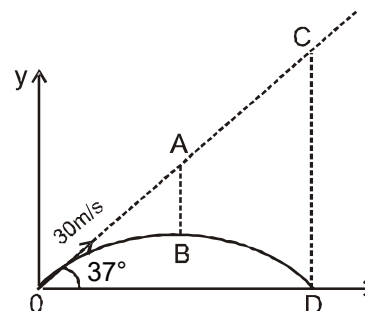


- (A) $\frac{mg}{2K}$ (B) $\frac{3mg}{K}$ (C) $\frac{3mg}{4K}$ (D) $\frac{3mg}{2K}$

17. Two swimmers 'A' & 'B', one located on one side and other on the another side of a river are situated at a distance 'D' from each other. Line joining them is making an angle θ with the direction perpendicular to flow. The speed of each swimmer with respect to still water is 'u' & speed of river flow is v_r . Both A & B start swimming at the same time in the direction parallel to line AB towards each other and they keep on swimming in same direction. Then find the time after which they will meet and also find the speed of river (v_r) in terms of 'u' so that the path of the two swimmers with respect to the ground becomes perpendicular to each other.

- (A) $\frac{D}{u}, 2u$ (B) $\frac{D}{2u}, 2u$
(C) $\frac{D}{2u}, u$ (D) $\frac{D}{u}, u$

18. A particle is projected with speed 30m/s at angle 37° with horizontal from ground as shown. AB and CD are parallel to y-axis and B is highest point of trajectory of particle. CD/AB is :-

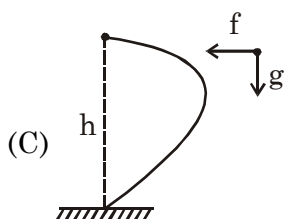
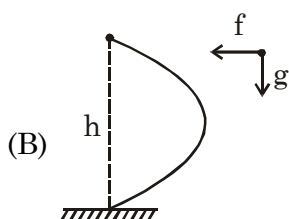
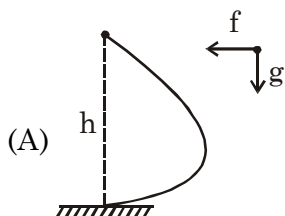


- (A) 3 (B) $\frac{3}{2}$
(C) 2 (D) 4

19. A body is projected at time $t = 0$ with a velocity u at an angle θ with the horizontal from ground. The horizontal and vertical components of its velocity will become equal at time $t > 0$.

- (A) If $0 \leq \theta \leq \frac{\pi}{4}$
(B) If $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$
(C) If $0 \leq \theta \leq \frac{\pi}{6}$
(D) for no value of θ

20. A stone is projected horizontally with speed v from a height h above ground. A horizontal wind is blowing in direction opposite to velocity of projection and gives the stone a constant horizontal acceleration f (in direction opposite to initial velocity). As a result the stone falls on ground at a point vertically below the point of projection. Its path would look like



(D) None of these

SECTION-II : (Maximum Marks: 20)

- This section contains **TEN** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

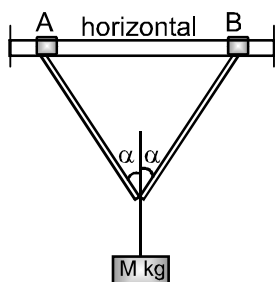
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

				+	
●	●	0	0	●	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

				-	
●	●	●	0	●	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
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9	9	9	9	9	9

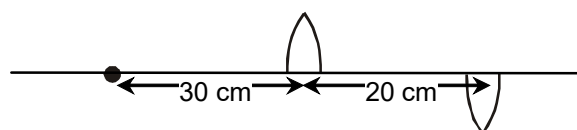
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. The system shown is in equilibrium in vertical plane. All the rods and the supports A & B are light and friction coefficient between supports and horizontal rod is μ . If maximum value of angle α is 37° then find the value of 4μ . Assume light rod exerts force only along the rod.



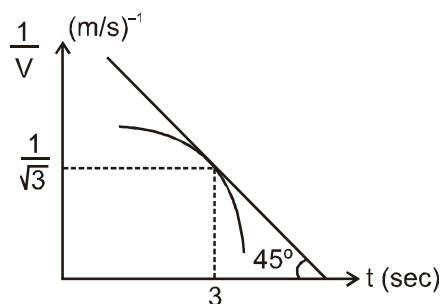
2. A particle of mass 2 kg experience a net force of 8N towards North. With respect to a reference frame S acceleration of the particle is 4m/s^2 towards 30° North of west. Another particle of mass 1 kg experience force of $2\sqrt{3}$ towards east. What should be magnitude acceleration of the particle with respect to frame S.

3. Thin convex lens of focal length f is split into two halves from its principal axis the separation between two lenses is 20cm the separation between nearest lens and object is 30 cm as shown in the figure. If image by nearest lens is formed at infinity. The distance (in cm) of the image of the object formed by second lens from its optical center is



4. An object of height 4cm is 20 cm away from concave mirror and is within focal length of the mirror. If the mirror is changed to a plane mirror, the image moves 15 cm closer to mirror. What is size of image in concave mirror (in cm).
5. For a prism kept in air it is found that for an angle of incidence 60° , the angle of refraction 'A', angle of deviation ' δ ' and angle of emergence 'e' become equal. Then the refractive index of the prism is :-

6. Two particles P_1 and P_2 are performing SHM along the same line about the same mean position. Initially they are at their positive extreme position. If the time period of each particle is 12 sec and the difference of their amplitudes is 12 cm then find the minimum time (in sec) after which the separation between the particles become 6 cm.
7. The graph shows the variation of $\frac{1}{V}$ (where V is the velocity of the particle) with respect to time. Then find the value of acceleration (in m/s^2) at $t = 3$ sec.



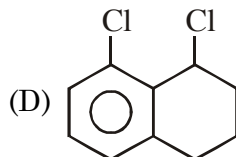
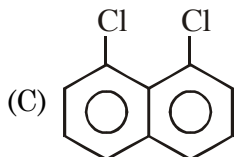
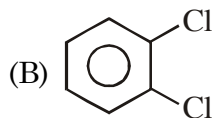
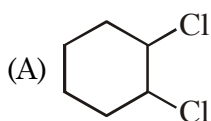
8. A body leaving a certain point O moves with an acceleration which is constant. At the end of the first second after it left O, its velocity is 1.5 m/sec. At the end of the sixth second after it left O the body stops momentarily and begins to move backwards. The distance traveled (in m) by the body before it stops momentarily is :
9. Two cars A and B are racing along straight line. Car A is leading, such that their relative velocity is directly proportional to the distance between the two cars. When the lead of car A is $\ell_1 = 10$ m, its running 10m/s faster than car B. The time car A will take to increase its lead to $\ell_2 = 20$ m from car B is t_1 then find value (in sec) of $10t_1$
10. Man A is sitting in a car moving with a speed of 12 m/s observes a man B in front of the car crossing perpendicularly a road of width 15 m in three seconds. Then the velocity of man B (in m/s) will be:

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PART 2 - CHEMISTRY

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Which of the following solid's anionic component has fractional bond order.
- (A) $\text{PCl}_5(\text{s})$ (B) $\text{N}_2\text{O}_5(\text{s})$
 (C) $\text{XeF}_6(\text{s})$ (D) $\text{P}_4(\text{s})$
2. Which of the following will show Geometrical isomerism.



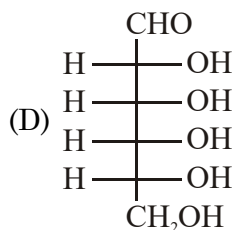
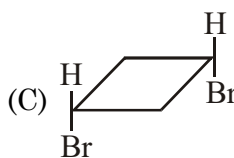
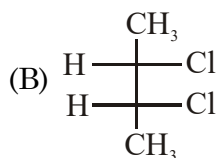
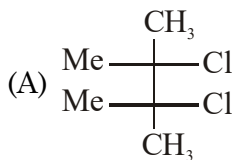
3. The solid A^+B^- crystallize in rock-salt structure. If $\left(\frac{r_{\text{A}^+}}{r_{\text{B}^-}}\right) = (\sqrt{2} - 1)$, the only

INCORRECT information is -

$$\left[\text{Take: } (\sqrt{2} - 1)^3 = 0.07, \frac{\pi}{\sqrt{2}} = 2.22 \right]$$

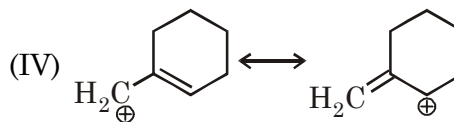
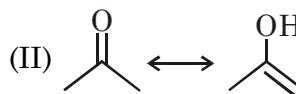
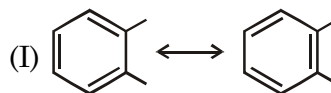
- (A) Anions are in contact with each other
 (B) Cations are in contact with each other
 (C) Edge length of unit cell, $a = 2\sqrt{2} \times r_{\text{B}^-}$
 (D) The packing fraction is nearly 0.79
4. The bond angle change due to back bonding will be observed in.
- (A) Be_2H_4 (B) AlCl_3
 (C) $(\text{CH}_3\text{O})_3\text{B}$ (D) BBr_3
5. Kjeldahl method is applicable for quantitative estimation of
- (A) Chlorine (B) Sulphur
 (C) Oxygen (D) Nitrogen
6. A liquid freezes at 7°C and boils at 77°C . If values of K_f and K_b of this liquid are 5.6 K-kg/mole and 2.5 K-kg/mole respectively, then ratio of molar latent heat of fusion (in kJ/mol) to molar latent heat of vaporisation (in kJ/mol) for the liquid is -
- (A) 1 : 1 (B) 7 : 2
 (C) 2 : 7 (D) 7 : 3

7. Which of the following order is **INCORRECT**?
- (A) $\text{CsCl} < \text{LiI}$ (Covalent character)
 (B) $\text{BaO} < \text{CaO}$ (Thermal stability)
 (C) $\text{AlCl}_3 < \text{MgCl}_2$ (Melting point)
 (D) $\text{RbHCO}_3 < \text{KHCO}_3$ (solubility in water)
8. Which one of the following is a meso compound -



9. A bottle of an aqueous H_2O_2 solution is labelled as 22.7 volumes and density of the solution is 1.5 gm/ml, then -
- (A) Molarity of solution is 1.39 M
 (B) Molality of solution is 2m
 (C) Molality of solution is 1.39 m
 (D) % w/w strength of solution is 6.8%
10. In which of the following reaction, the hybridization of underlined atom is changed from reactant to product.
- (A) $\underline{\text{B}}_2\text{H}_6 + 2\text{CO} \longrightarrow$
 (B) $\underline{\text{B}}_2\text{H}_6 + 2\text{LiH} \longrightarrow$
 (C) $\underline{\text{B}}_2\text{H}_6 + 2\text{C}_2\text{H}_5\text{NH}_2 \longrightarrow$
 (D) None of these

11. Which of the following pairs are resonance structures of each other?

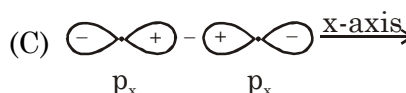
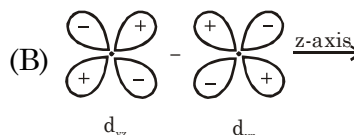
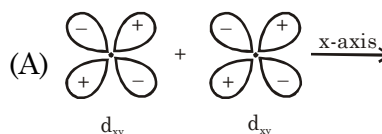


- (A) I, II, III (B) I, IV
 (C) II, III (D) I, III, IV

12. If 40.6 g of a hydrocarbon contains 4.2×10^{23} molecules, then the hydrocarbon can be [$N_A = 6 \times 10^{23}$]

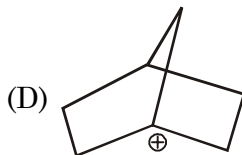
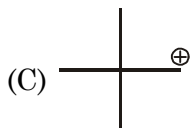
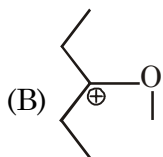
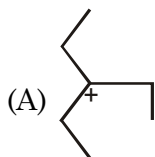
- (A) C_2H_2 (B) C_3H_8
 (C) C_4H_{10} (D) C_2H_6

13. Which of the following overlap will form a molecular orbital having Gerade symmetry?



- (D) Both (A) & (B)

14. Which of following is most stable carbocation?



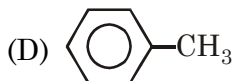
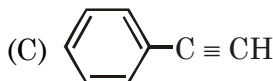
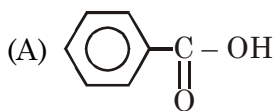
15. 1.5 gm of an unknown gas at 127°C occupies the same volume as 9.6 gm ozone gas at 27°C, at the same pressure. The molar mass of unknown gas is -

- (A) 10 gm/mol (B) 5 gm/mol
(C) 15 gm/mol (D) 20 gm/mol

16. Which of the following specie does not have a bridging atom between two atoms of same kind.

- (A) $\text{N}_2\text{O}_5(\text{g})$
(B) S_3O_9
(C) Planar Dimeric BeCl_2
(D) Cl_2O_6

17. Which of the following compound is most acidic?



18. An aqueous solution contains a mixture of $\text{Ca}(\text{NO}_3)_2$ and $\text{Al}(\text{NO}_3)_3$. A sample of this aq. solution is 2M in $\text{Ca}(\text{NO}_3)_2$ and 3m in $\text{Al}(\text{NO}_3)_3$. What should be the density of solution such that the molality and molarity of NO_3^- ion becomes numerically same?

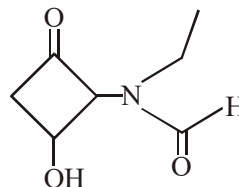
[Molar masses : $\text{Ca}(\text{NO}_3)_2 = 164 \text{ g/mol}$,
 $\text{Al}(\text{NO}_3)_3 = 213 \text{ g/mol}$]

- (A) 1.0 gm/ml
(B) 1.967 gm/ml
(C) 1.310 gm/ml
(D) 1.918 gm/ml

19. Which of the following has odd electron in the orbital having minimum percentage 'p' character.

- (A) $\dot{\text{C}}\text{ClO}_3$ (B) $\dot{\text{C}}\text{F}_3$
(C) $:\text{CCl}_2$ (D) NO_2

20. The principle functional group in following compound for IUPAC naming.



- (A) Aldehyde (B) Ketone
(C) Amine (D) Amide

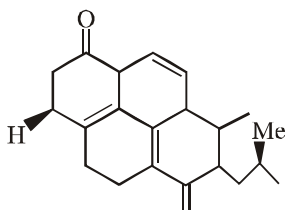
SECTION-II : (Maximum Marks: 20)

- This section contains **TEN** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

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7. 100 ml aqueous solution containing equimolar mixture of Ca(OH)_2 and Al(OH)_3 requires 0.5 litre of 4M HCl for complete neutralisation. Molarity of Ca(OH)_2 in the original solution is (Assume Ca(OH)_2 and Al(OH)_3 completely ionized in solution)
8. Number of specie(s) that have tendency to dimerize among the following.
 $\dot{\text{C}}\text{H}_3$, BeH_2 , BCl_3 , AlCl_3 , ClO_2 , ICl_3 , CH_3COOH in Benzene, SiCl_4
9. How many chiral carbon atoms are present in the following compound.



10. In CaF_2 -type lattice, if
 x = nearest distance between Ca^{2+} & F^- ions
 y = nearest distance between two F^- ions
 z = nearest distance between two Ca^{2+} ions.
 then find value of the expression.

$$\left[\sqrt{6} \left(\frac{xz}{y^2} \right) \right]$$

ALLEN
PART 3 - MATHEMATICS

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The number of value(s) of 'a' for which the equations $x^2 - 3x + 2 = 0$, $x^2 - 4x + a = 0$ and $x^2 + 2x - a = 0$ possess exactly one common root, is :-
 (A) 0 (B) 1
 (C) 2 (D) Infinite
2. Let α and β are two distinct real roots of the quadratic equation $(2a - 1)x^2 + x - 1 = 0$. If $\alpha < 2 < \beta$ then 'a' cannot take the value
 (A) $\frac{13}{32}$ (B) $\frac{15}{32}$ (C) $\frac{7}{16}$ (D) $\frac{5}{8}$
3. If $ax^9 + bx^8 + 1$ is divisible by $x^2 - x - 1$ then value of 'a' is :-
 (A) 1 (B) 21 (C) 11 (D) 10

4. $\lim_{x \rightarrow 2} \frac{\sin(x^2 - 5x + 6)}{x^2 - 3x + 2}$ is equal to :-
 (A) 2 (B) 1 (C) -1 (D) 0
5. Number of real solution(s) of the equation $\sin^{-1}(\sin x) = 1 - |x|$ is :-
 (A) 1 (B) 2
 (C) 3 (D) infinite
6. Value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right)$ is :-
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) π
7. The value of $\sin(\cot^{-1}3 + \cot^{-1}5)$ is :-
 (A) $\frac{7}{\sqrt{65}}$ (B) $\frac{4}{7}$ (C) $\frac{4}{\sqrt{65}}$ (D) $\frac{3}{7}$
8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $3f(x) + 2f(-x) = x^2 - x \forall x \in \mathbb{R}$ then $f(x)$ is :-
 (A) one-one function
 (B) many-one function
 (C) odd function
 (D) invertible function
9. If the equation $|x^2 - 2x| = \lambda$ has exactly six different real solutions then value of λ can be :-
 (A) 0 (B) $\frac{4}{7}$
 (C) 1 (D) $\frac{7}{4}$

10. The domain of the function

$$f(x) = \frac{x^3}{\sqrt{\sin^{-1}(\ln x) - \cos^{-1}(\ln x)}} \text{ is :-}$$

(where $\ln x = \log_e x$)

(A) $\left(e^{\frac{1}{2}}, e^{\frac{1}{\sqrt{2}}}\right)$ (B) $\left(e^{\frac{1}{\sqrt{2}}}, e\right)$

(C) $\left[e^{\frac{1}{2}}, e^{\frac{1}{\sqrt{2}}}\right]$ (D) $\left[e^{\frac{1}{\sqrt{2}}}, e\right]$

11. Let $f(x) = \begin{cases} x+4 & 0 \leq x \leq 4 \\ -3x-2 & -2 < x < 0 \end{cases}$ and

$$g(x) = \begin{cases} 2x+2 & x \geq -1 \\ 3x^2+3 & x < -1 \end{cases}, \text{ then the value of}$$

$g(f(1)) + g(f(-1))$ is :-

- (A) 8 (B) -6
(C) 16 (D) 12

12. Let $f(x) = x^8 - 6x^7 + 5x^6 + x^4 - 5x^3 + x^2 - 3x + 3 \forall x \in \mathbb{R}$ then $f(5)$ is equal to :-

- (A) 0 (B) 3
(C) 13 (D) 10

13. Let $L = \lim_{x \rightarrow \infty} \left(ax^2 - \frac{1+x^4}{1-x^2} \right)$ if 'L' is a non zero

finite number then choose the correct option

- (A) $a = 0$ (B) $a = -1$
(C) $L = -2$ (D) $L = -1$

14. Let $f: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, -\frac{\pi}{4}\right]$ be a function defined

as $f(x) = \tan^{-1}(2x - x^2 - 2)$ then function $f(x)$ is

- (A) one-one and onto
(B) one-one and into
(C) many one and onto
(D) many one and into

15. Let $L = \lim_{x \rightarrow \frac{\pi}{6}} \operatorname{sgn} \left(\cos 2x - 2 \sin x + \frac{1}{2} \right)$

(where $\operatorname{sgn}()$ is the signum function). Choose the correct option

- (A) $L = 0$ (B) $L = -1$
(C) $L = 1$ (D) 'L' does not exist

16. If the complete set of values of the

$$\text{expression } y = \frac{11x^2 + 12x + 6}{x^2 + 4x + 2} \text{ is}$$

$(-\infty, a] \cup [b, \infty)$ then the value of 'a + b' is

- (A) 2 (B) -2 (C) 8 (D) -8

17. If $\lim_{x \rightarrow 0} \frac{\tan \left(\frac{x+x^2}{a^4} \right) (e^x - 1)^2}{\ln \left(1 + \frac{x^3}{2} \right)} = \frac{2}{16}$, then value

of 'a' can be

- (A) 3 (B) -3 (C) 2 (D) $\frac{1}{2}$

18. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = 4 + x^3 + 2^x \forall x \in \mathbb{R}$. If 'g' be the inverse function of 'f' then $g(7)$ is

- (A) $3 + \log_e 2$ (B) 1
(C) 4 (D) 2

19. If $f(x) = \{x+1\} + \{x+3\} + \{x+5\} + \dots + \{x+199\}$

then value of $\left[f(\sqrt{3}) \right]$ is :-

(Where $\{.\}$ is fractional part function and $[.]$ is greatest integer function)

- (A) 10732 (B) 73
(C) 173 (D) 100

20. Let $f : (0, \infty) - \{1\} \rightarrow \mathbb{R}$ such that

$$f(x) = \log_{x^4} x^3 \text{ then } f(x) \text{ is}$$

- (A) one one and onto
(B) many one and into
(C) many one and onto
(D) one one and into

SECTION-II : (Maximum Marks: 20)

- This section contains **TEN** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Let $a(x) = \alpha^x + \beta^x \forall x \in \mathbb{R}$, where α and β are roots of the equation $x^2 + 3x + 1 = 0$.
If $a(8) + a(6) = \lambda a(7)$ then value of $\frac{|\lambda|}{5}$ is
2. The maximum value of $f(x) = \log_5(3\sin x + 4\cos x + 20) \forall x \in \mathbb{R}$ is
3. If equation $\tan^{-1}(x^2 + 5x + 3) + \cot^{-1}(2x + a) = \frac{\pi}{2}$ has a real solution then minimum value of 'a' is
4. If the function $f(x) = \frac{2^x - 1}{x^a(2^x + 1)}$ is an even function then smallest positive integral value of 'a' is
5. If the number of solution(s) of the equation $6\{x\} - 3[x] = 2x$ is 'N' then value of 'N + 2' is (where $\{ \}$ fractional part function and $[\cdot]$ greatest integer function)
6. Let $f(x) = \begin{cases} x^2 & -1 \leq x < 2 \\ x+2 & 2 \leq x \leq 3 \end{cases}$ and $g(x) = \begin{cases} x+1 & x \leq 1 \\ ax+1 & 1 < x \leq 2 \end{cases}$ if $\lim_{x \rightarrow 1} g(f(x))$ exists and finite then value of 'a' is
7. Let $f(x)$ be a function which satisfies the relation $f(x+1) + f(2x+2) + f(1-x) + f(2+x) = x+1 \forall x \in \mathbb{R}$ then value of $|f(0)|$ is
8. If the value of $\sin^{-1}(\sin 8) + \cos^{-1}(\cos 6) + \tan^{-1}(\tan 9) = a\pi + b$ where a and b are integers then the value of 'a - b' is
9. Number of integers in the range of the function $f(x) = \sin^{-1}x + |\sin^{-1}x| + \sin^{-1}|x|$ is
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an even function satisfying $f(x) = f(x+6) \forall x \in \mathbb{R}$ such that $f(x) = \begin{cases} 4x & 0 \leq x < 1 \\ 5-x & 1 \leq x \leq 3 \end{cases}$. The value of $f(8) + f(-3)$ is

